

CSE-355 (Spring 2026)

HW 01

Homework 1

Total Marks: 100

Deadline: Feb 22, 2026 at 11:55pm

General Instructions

- There are a total of 10 questions and each question carries 10 marks. (some are easier than the others and this has been done on purpose)
- This assignment must be completed and submitted individually. Group submissions are NOT allowed
- Simply dividing questions among peers or sharing written solutions is strictly prohibited and defeats the purpose of the assignment
- Every student is expected to develop a complete and independent understanding of all questions
- You may use online resources, including Generative AI tools such as ChatGPT, strictly for learning and clarification purposes. However, copying solutions, generating answers without understanding them, or submitting AI-produced content verbatim is considered academic misconduct
- If you have any confusion, please post your questions on the Whatsapp Group of the course or contact the TA or the Instructor directly

Submission Instructions

- All solutions must be typed and submitted in PDF format. **Handwritten submissions will NOT be accepted under any circumstances.**
- Typesetting your solutions in **LaTeX** is strongly encouraged and preferred. However, you can also use MS Word or Google Docs, etc.
- Submissions must be made on LMS before the stated deadline. Email submissions will NOT be accepted. The deadline will not be extended without imposing a penalty for late submission.
- Each student must submit their own assignment individually. The first page must clearly include your full name and student ID.

Question 1

Suppose Alice wants to send a confidential message to Bob. She does not want anyone else to see what she sent. Alice also wants Bob to be sure it really came from her.

How can we meet all the requirements? Explain by specifying all the actions taken by both Alice and Bob. Feel free to simply draw a block diagram.

Question 2

Bob receives a message that claims to be signed by Alice. He tries to verify it using Alice's public key, but the verification fails.

- a) What does this outcome mean?
- b) Based on this, what possible issues or attacks should Bob suspect?

Question 3

a) Alice comes up with her own digital signature scheme. She writes a message and encrypts that message using her Private key. She then sends this message and the encrypted message both to Bob. Can Bob be sure that this message came from Alice?

b) Alice modifies this scheme. She writes a message and encrypts that with her private key (as in the previous part). She then hashes the encrypted message and sends this hash along with the original message to Bob. Can Bob be sure that this message is from Alice?

Question 4

In class we learnt that a symmetric key is often used for communication *in-bulk* i.e., when there is a lot of data that must be transferred. This is done because symmetric keys operate a lot faster than asymmetric keys.

a) Assume that Alice and Bob share a secret, symmetric key. Devise a scheme with which Alice can send a plain-text message to Bob and ensure that Bob can verify the authenticity of this message.

Assume that Alice and Bob must share a LOT of data and asymmetric keys are not viable.

b) Consider a scenario in which Alice wishes to relay to Bob the message "Give me 3.50\$". A malicious actor, Kevin, duplicates this message and sends it twice to Bob (this is called a replay attack). According to the scheme you devised in the previous part, what will the outcome be? Further, devise a scheme in which duplicated messages are rejected.

Question 5

In a Bitcoin blockchain, a transaction claims "Alice pays 0.3 BTC to Bob."

- a) What checks must be performed on Alice's digital signature before this transaction is considered valid?
- b) Could Alice later deny sending it? Why or why not?

Question 6

Consider a blockchain with three blocks. Block 1 contains transactions and has a hash $H1 = \text{ABC123}$. Block 2 contains its own transactions and includes $H1$ in its data before being hashed to produce $H2 = \text{DEF456}$. Block 3 does the same, resulting in $H3 = \text{GHI789}$.

- a) If a transaction in Block 1 is altered, explain why merely recalculating its hash to $H1'$ is insufficient to hide the tampering.
- b) What must an attacker do to make the altered chain appear valid to a node synchronizing from the latest block? Why does Proof-of-Work make this practically very difficult?

Question 7

Consider a simple encryption procedure using a public/private key pair. You have the public key as the integer $p=7$ and the “secret” private key as the integer $s=3$.

The method to encrypt an integer m is: $c = m^p \bmod 33$

And the method to decrypt an encrypted integer is: $m = c^s \bmod 33$

Assume that the input characters have the following numerical values:

$a/A = 0, b/B = 1, c/C = 2, \dots, z/Z = 25$.

- a) Encrypt the word ‘SurpriseQuiz’ and write down the sequence of numbers.
- b) Decrypt the sequence “1 2 13 29 20 2 7 20 21 16 8 16 13 28”. Write down the characters and what you think of them.

Now let's consider a hashing function $h(x) = \lfloor e^x + 100 \rfloor \bmod 32$ where $x \in [0, 50]$.

As $h(x)$ is defined only for one input, we will modify it to take in lists of numbers such as “31 23 4 2 13”. We define that $h(31\ 23\ 4\ 2\ 13)$ will be $h((31 + 23 + 4 + 2 + 13) \bmod 50)$. The idea is that every number in the list makes a contribution. Suppose Alice has the public key $p = 7$ and private key $s = 3$. Data is sent in the form “ $m_1 m_2 m_3 \dots m_n$ sig” where the m_i are numbers from the message and “sig” is the **private** key (encryption) of the hash of the message. Note that this is different from the previous parts.

- c) Suppose Alice wants to transmit the word “SaveMe”. What is the final message that Alice transmits?
- d) You receive a message “12 20 18 19 0 5 0 17 14 2 10 18 1 0 2 7 4 18 7 14 2 10 18 5”. It is known that the scheme in part (c) is used. Is this message from Alice? Why or why not? If it is a message from Alice then write the message.

Question 8

- a) What is the fundamental difference between Symmetric and Asymmetric (Public Key) cryptography?
- b) For the following goals, state whether symmetric or asymmetric cryptography is typically used. Also state your reasoning: - i) Encrypting a large file for storage. - ii) Proving your identity to a website. - iii) Establishing a secure connection to your bank's server.

Question 9

A simple hash function is defined as $H(x) = (5x + 3) \bmod 16$.

- a)** Calculate the hash of the following numbers: $H(7)$, $H(23)$, $H(39)$.
- b)** What do you observe about the results for $H(7)$ and $H(23)$? Explain why this happens, relating it to the properties of the modulo operation.
- c)** Based on this, what is a major weakness of this hash function?

Question 10

We briefly talked about Proof-of-Stake (PoS) as an alternative to Proof-of-Work (PoW).

- a)** In PoW, miners expend energy. What do validators in a PoS system “stake” or put at risk to participate in block creation?
- b)** Read up on the advantages and criticisms of PoS systems, and mention one for each here.